LATIHAN SOAL PROBABILITAS

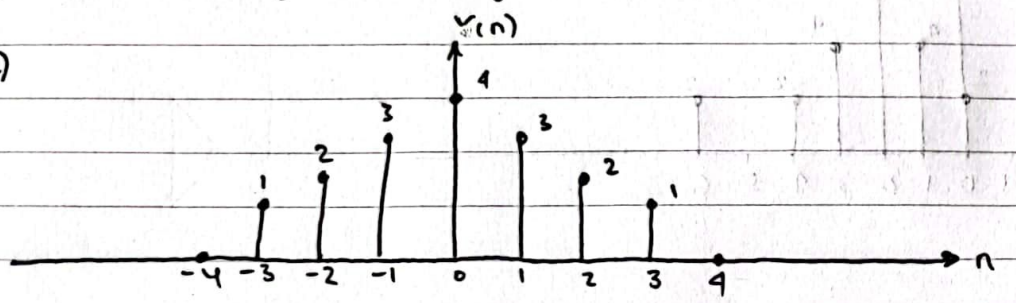
No. 7/11/22

Date

211344013 - ELSADANITA MAURA SITORUS

① Tentukan daya dan energi dari sinyal berikut :

(a)



Energy signal →

→ $E = \sum_{n=-4}^{4} V^2(n)$

$= 0^2 + 1^2 + 2^2 + 3^2 + 4^2 + 3^2 + 2^2 + 1^2 + 0^2 = 44$ //

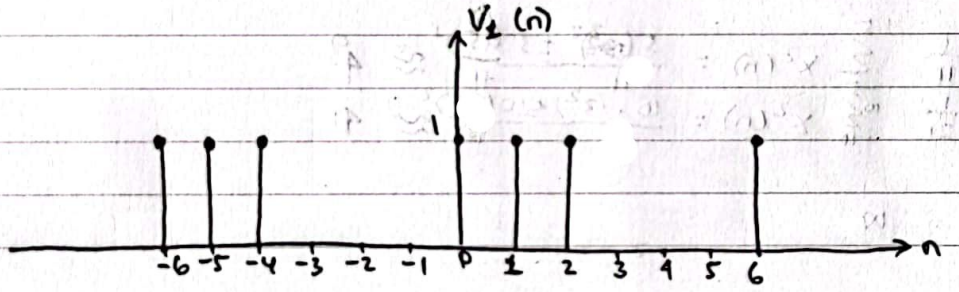
→ $P \rightarrow \frac{1}{21} \sum_{-10}^{10} V^2(n) = 44/21$ //

∴ Maka $E = 44$

$P = \infty$

→ $P \rightarrow \frac{1}{11} \sum_{-5}^{5} V^2(n) = 44/11$ //

(b)



→ $N = 5 \rightarrow P = \frac{1}{11} \sum_{n=-5}^{5} V^2(n) = \frac{5 \cdot 1^2}{11} = \frac{5}{11} \approx 0{,}45$

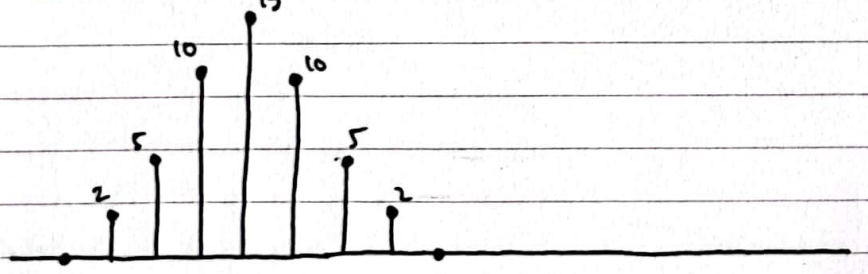
→ $N = 10 \rightarrow P = \frac{1}{21} \sum_{n=-10}^{10} V^2(n) = \frac{10 \cdot 1^2}{21} = \frac{10}{21} \approx 0{,}47$

∴ Maka $P = 0{,}5$ //

$E = \sum_{n=\infty}^{\infty} V^2(n) = \infty$ //

② Autokorelasi dari sinyal soal ①

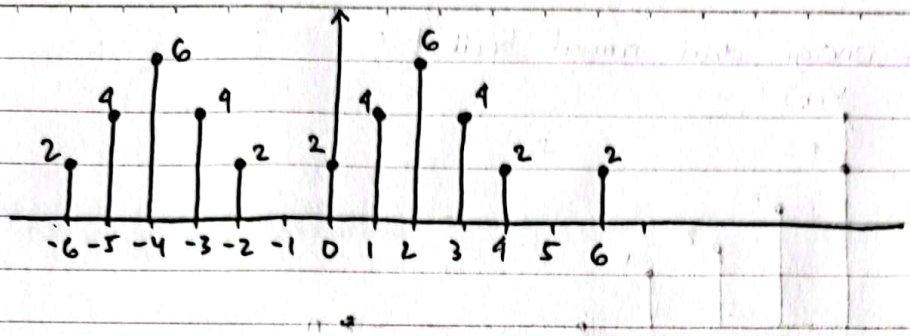
(a)



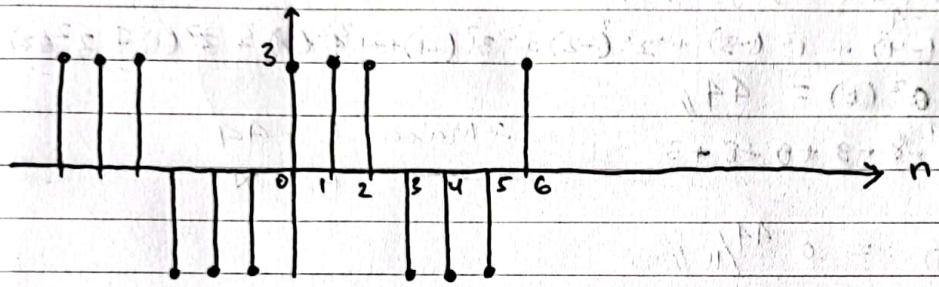
No.

Date

(b)



(3) Tentukan daya dan energi dari sinyal berikut :



⇒ $N = 5 \rightarrow P = \frac{1}{11} \sum_{-5}^{5} x^2(n) = \frac{5(3^2) + 5(-3)^2}{11} = 9$

$N = 10 \rightarrow P = \frac{1}{21} \sum_{-10}^{10} x^2(n) = \frac{10 \cdot (3^2) + 10(-3)^2}{21} = 9$

⇒ $E = \sum_{n=\infty}^{\infty} v^2(n) = \infty$

